Critical Questions for Big Data

CT0540 - Social network analysis

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Introduction

- Big Data is a poor term
- Used to refer to data sets large enough to require supercomputers (not valid anymore) [Manovich, 2011]
- It is true that the quantities of data now available are often quite large..
- .. but that is not the defining characteristic of Big Data

Introduction

- Some of the data encompassed by Big Data are not as large as earlier data sets that were not considered as such (e.g. census data)

Introduction

- Big Data is about a capacity to **search**, **aggregate**, and **cross-reference** large data sets
- Two-fold interpretation:
  - As a powerful tool to address societal issues
  - As a troubling manifestation of Big Brother, enabling invasions of privacy, decreased civil freedoms, and increased state and corporate control

Introduction

- As with all socio-technical phenomena, we have nuances
- We observe:
- Increased automation of data collection and analysis
- Algorithms that can extract and illustrate large-scale patterns in human behavior
- But which systems are driving and regulating these practices?

Introduction

- Social systems are regulated by four forces [Lessig, 1999]:
  - Market
  - Law
  - Social norms
  - Architecture (code)
- These forces for Big Data are at odds

Introduction

- Market: Big Data = opportunity
- Law: Big Data = concerns about privacy
- Social norms and code:
  - Personalization = more relevant information
  - But comes with ethical questions and fragment the public

Critical Questions for Big Data

The authors critically interrogate the assumptions and biases of Big Data:

- What does all this data mean?
- Who gets access to what data?
- How is data analysis deployed?
- To what ends?

Critical Questions for Big Data

- They offer **six provocations** on Big Data in social media context
- Hard questions with no easy answers

1. Big Data changes the definition of knowledge

- Big Data not only refers to very large data sets and the tools and procedures used to manipulate and analyze them, but also to a computational turn in thought and research
- Big Data has emerged a system of knowledge that has the power to inform how we understand human networks and community

1. Big Data changes the definition of knowledge

- The specialized tools of Big Data have their own inbuilt limitations and restrictions
- Example:
  - Twitter and Facebook offer very poor archiving and search functions
  - Researchers are much more likely to focus on something in the present or immediate past because of the difficulty/impossibility of accessing older data

2. Claims to objectivity and accuracy are misleading

- All researchers are interpreters of data
- A model may be mathematically sound, an experiment may seem valid, but as soon as a researcher seeks to understand what it means, the process of interpretation has begun
  - Not all interpretations are created equal, but also not all numbers are neutral

2. Claims to objectivity and accuracy are misleading

- The design decisions that determine what will be measured also stem from interpretation
- Example: Social media data
  - **Data cleaning** process
  - Making decisions about what attributes and variables will be counted, and which will be ignored
  - This process is subjective

2. Claims to objectivity and accuracy are misleading

- There is also the issue of data errors
- Large data sets from Internet sources may be unreliable
- These errors and gaps are magnified when multiple data sets are used together
- Important to ask critical questions about the collection of data and try to account for any biases

2. Claims to objectivity and accuracy are misleading

What should we do?

- Understanding the properties and limits of a data set, regardless of its size
  - A data set may have many millions of pieces of data, but this does not mean it is random or representative
  - To make statistical claims about a data set, we need to know where data is coming from and account for the weaknesses in that data
- Accounting for the biases in our interpretation of the data
  - Interpretation is at the center of data analysis and is subject to limitation and bias
  - Without those biases and limitations being understood and outlined, misinterpretation is the result

3. Bigger data are not always better data

- There are many ways to assess the validity of a work
  - Survey researchers focus on sampling mechanisms and question bias
  - Quantitative researchers weigh up statistical significance
- Having large quantities of data does not mean that methodological issues are not relevant
- Twitter provides an example in the context of a statistical analysis

3. Bigger data are not always better data

- It is relatively easy to obtain Twitter data
- Scholars have used Twitter to examine a wide variety of topics, patterns, and dynamics
- Many scholars discuss the limitations of Twitter data in their publications, but..
- .. the public discourse around such research tends to focus on the raw number of tweets available
- What's the problem?

3. Bigger data are not always better data

- Twitter does not represent **all people**
- **Twitter users** are a very particular subset
- The population using Twitter representative of the global population
- Accounts and users are equivalent
  - Multiple accounts
  - Accounts used by multiple people
  - No account, simple access
  - Bots

3. Bigger data are not always better data

- The notion of an 'active' account is problematic
  - Some users post content frequently
  - Others participate as 'listeners'
  - According to Twitter, 40% of active users sign in just to listen [Source]

3. Bigger data are not always better data

- Big Data and whole data are not the same
- Without taking into account the sample of a data set, the size of the data set is meaningless
- Example:
  - A researcher wants to understand the topical frequency of tweets
  - If Twitter removes all tweets that contain problematic words or content from the stream, the topical frequency would be inaccurate
  - The data is skewed from the beginning

3. Bigger data are not always better data

- Often the source is uncertain
- Facebook makes a fraction of its material available to the public through its APIs
  - Posts that a user chose to make private or 'protected' are not included
  - Few researchers have access to the whole data
  - Facebook Graph API return approximately 600 ranked, published posts per year
  - It is not not clear what posts are included or how they are sampled
  - Without knowing, it is difficult for researchers to make claims about the quality of the data that they are analyzing
  - Are the data representative of all posts? No

4. Taken out of context, Big Data loses its meaning

- Social media data are largely used to make claims about social networks
- Yet, the relations displayed through social media are not necessarily equivalent to the sociograms and networks investigated by sociologists
- The ability to represent relationships between people as a graph does not mean that they convey equivalent information

4. Taken out of context, Big Data loses its meaning

- Historically, data about people's relationships has been collected through surveys, interviews, observations, and experiments to describe people's **personal networks**
- Big Data introduces two types of social networks:
  - **Articulated networks**: e.g., Friends/Followers lists – can include friends, colleagues, acquaintances, celebrities, etc..
  - **Behavioral networks**: derived from communication patterns and social media interactions

4. Taken out of context, Big Data loses its meaning

- Behavioral and articulated networks have great value to researchers
- But they are not necessarily equivalent to personal networks
  - Example: Measuring tie strength through frequency or public articulation can be a mistake
- Not every connection is equivalent to every other connection
- Frequency of contact does not indicate strength of relationship

5. Just because it is accessible does not make it ethical

- In 2006, a Harvard-based research group started gathering the profiles of 1,700 college-based Facebook users to study how their interests and friendships changed over time [Lewis et al., 2008]
- These supposedly anonymous data were released to the world, allowing other researchers to explore and analyze them
- What other researchers quickly discovered was that it was possible to de-anonymize parts of the data set: compromising the privacy of students, none of whom were aware their data were being collected [Zimmer, 2008]

5. Just because it is accessible does not make it ethical

Lewis et al detail some of the steps taken to protect the privacy of the subjects:

- All identifying information was deleted or encoded immediately after the data were downloaded
- The roster of student names and identification numbers is maintained on a secure local server accessible only by the authors of this study. This roster will be destroyed immediately after the last wave of data is processed
- The complete set of cultural taste labels provides a kind of "cultural fingerprint" for many students, and so these labels will be released only after a substantial delay in order to ensure that students' identities remain anonymous
- In order to access any part of the dataset, prospective users must read and electronically sign an user agreement

5. Just because it is accessible does not make it ethical

Some of the concerns raised by Zimmer:

- One might think "all identifying information" has been deleted, but often random bits of our **data trail that alone seem anonymous can be pieced together**, possibly exposing clues to our identity.
- The fact that the dataset includes each subjects' gender, race, ethnicity, hometown state, and major makes it increasingly possibility that individuals could be identified.
- If the data reveals that student #746 is a white Bulgarian male from Montana, majoring in East Asian Studies, there probably aren't that many who fit such a description. Unlikely, but not bullet-proof.

5. Just because it is accessible does not make it ethical

Some of the concerns raised by Zimmer:

- Instituting a "substantial delay" before releasing this personal data does little to mitigate the privacy fears...it only delays them.
- Delaying the release of this data to 2011 is not only arbitrary, but much too short.
- A better tactic would be to gain the consent of the subject before releasing the data, or simply not releasing it at all to provide the fullest privacy protection of the subjects.

5. Just because it is accessible does not make it ethical

Some of the concerns raised by Zimmer:

- Since I first saw the press release for this dataset, I've been bothered by the description of the date as "approximately 1,700 FB profiles by an entire cohort of students at an anonymous, northeastern American university."
- The source university loses full anonymity since it is identified as being in the northeastern US
- According to the codebook, this is a private, co-ed institution, whose class of 2009 initially had 1640 students in it.

5. Just because it is accessible does not make it ethical

Some of the concerns raised by Zimmer:

- A quick search for schools reveals there are only 7 private, co-ed colleges in New England states with total undergraduate populations between 5000 and 7500 students (a likely range if there were 1640 in the 2006 freshman class): Tufts University, Suffolk University, Yale University, University of Hartford, Quinnipiac University, Brown University, and Harvard College
- Is one of these the source?This might prove easy to discover, given the uniqueness of some of the subjects.
- Based on the codebook, the dataset includes only one self-identified Albanian, one Iranian, one Malaysian, one Nepali, and other solitary ethnicities.

5. Just because it is accessible does not make it ethical

Some of the concerns raised by Zimmer:

- If we can isolate one of these people in the dataset, combined with the subject's gender, home state, and major, it probably wouldn't be that hard to discover who it is.
- Keying off these unique data elements will provide a possible path to identifying the school, and potentially many more individuals in the dataset.
- This would have been much harder if I didn't know it was a private school in the northeast us. Or if they took a random sample of the dataset, and didn't tell me that actual number of students in the cohort.

5. Just because it is accessible does not make it ethical

The authors provided a response to Zimmer' concerns:

- *Kaufman*: What might hackers want to do with this information, assuming they could crack the data and 'see' these people's Facebook info? Couldn't they do this just as easily via Facebook itself? Our dataset contains almost no information that isn't on Facebook. (Privacy filters obviously aren't much of an obstacle to those who want to get around them.)
- *Halavais*: The data is already there, this is merely (!) the collection of that data. Or to put it another way, [..] Facebook users are intending to share with someone (if not the researchers).

5. Just because it is accessible does not make it ethical

How Zimmer reacted:

- One, Kaufman's mention of "hackers" and focusing on what they might "do" with this information exposes a focus on "harms" when it comes to privacy concerns.
- One doesn't need to be a victim of hacking, or have a tangible harm take place, in order for there to be concerns over the privacy of one's personal information.
- Privacy is about dignity as much as about informational harm by some evil agent. As Havalais points out later in his comment, none of the subjects in this dataset consented to having their personal information used in a research study. Don't they have a right to some control over their information?

5. Just because it is accessible does not make it ethical

How Zimmer reacted:

- Just because users post information on Facebook doesn't mean they intend for it to be scraped, aggregated, coded, disected, and distributed.
- Creating a Facebook account and posting information on the social networking site is a decision made with the intent to engage in a social community, to connect with people, share ideas and thoughts, communicate, be human.
- Just because some of the profile information is publicly avaiable (either consciously by the user, or due to a failure to adjust the default privacy settings), doesn't mean there are no expectations of privacy with the data.

5. Just because it is accessible does not make it ethical

How Zimmer reacted:

- First, the researchers received an official roster of each freshman at the college, along with their university e-mail address. This allowed them to easily and systematically search for each student in the cohort.
- Some Facebook users might have restricted their accounts to only people from their own school. But since the researchers used research assistants from that school to access to account information, that restricted data has been published outside of those boundaries.

5. Just because it is accessible does not make it ethical

- The case raised difficult issues for scholars:
  - What is the status of so-called 'public' data on social media sites?
  - Can it simply be used, without requesting permission?
  - What constitutes best ethical practice for researchers?

5. Just because it is accessible does not make it ethical

- Research ethics committees emerged in the 1970s to oversee research on human subjects
  - Provide a framework for evaluating the ethics of a particular line of research
  - Make certain that checks and balances are put into place to protect subjects (informed consent, privacy protection)

5. Just because it is accessible does not make it ethical

- Very little is understood about the ethical implications underpinning the Big Data phenomenon:
  - Should someone be included as a part of a large aggregate of data?
  - What if someone's 'public' blog post is taken out of context and analyzed in a way that the author never imagined?
  - What does it mean for someone to be spotlighted or to be analyzed without knowing it?
  - Who is responsible for making certain that individuals and communities are not hurt by the research process?
  - What does informed consent look like?

5. Just because it is accessible does not make it ethical

- It may be unreasonable to ask researchers to obtain consent from every person who posts content on social media
- But just because content is publicly accessible does not mean that it was meant to be consumed by just anyone
- Researchers must keep asking themselves – and their colleagues – about the ethics of their data collection, analysis, and publication

5. Just because it is accessible does not make it ethical

- Social media users data were created in highly context-sensitive spaces
- It is entirely possible that some users would not give permission for their data to be used elsewhere
- Many are not aware of the multiplicity of agents and algorithms currently gathering and storing their data for future use
- Users are not necessarily aware of all the multiple uses, profits, and other gains that come from information they have posted
- Data may be public (or semi-public) but this does not simplistically equate with full permission being given for all uses

6. Limited access to Big Data creates new digital divides

- While the explosion of research using data sets from social media sources would suggest that access is straightforward, it is anything but
- Only social media companies have access to really large social data
- Researchers working for these companies will have access to data that the rest of the scholarly community will not
- Some companies restrict access to their data entirely; others sell the privilege of access for a fee; and others offer small data sets to university-based researchers
- This produces considerable inequality in the system

6. Limited access to Big Data creates new digital divides

- In addition to questions of access, there are questions of skills
- Wrangling APIs, scraping, and analyzing big swathes of data is a skill set generally restricted to those with a computational background
- But computer scientists and social scientists both have valuable perspectives to offer
- This is also a gendered division. Most researchers who have computational skills at the present moment are male and **who is asking the questions determines which questions are asked**

6. Limited access to Big Data creates new digital divides

- Large data companies have no responsibility to make their data available
- Researchers with access to proprietary data sets might be less inclined to choose questions that are contentious to a social media company if they think it may result in their access being cut
- The effects on the kinds of research questions that can be asked are something to be considered

Notes

This presentation is entirely based on Boyd, D., & Crawford, K. (2012). Critical questions for big data: Provocations for a cultural, technological, and scholarly phenomenon. Information, communication & society, 15(5), 662-679.